EL9: Documentation

Quarto

Reading

- [1, ch. 28-29] introduces Quarto with good examples

For reference

quarto.org

Levels of documentation

Documentation can exist at several levels in a project.

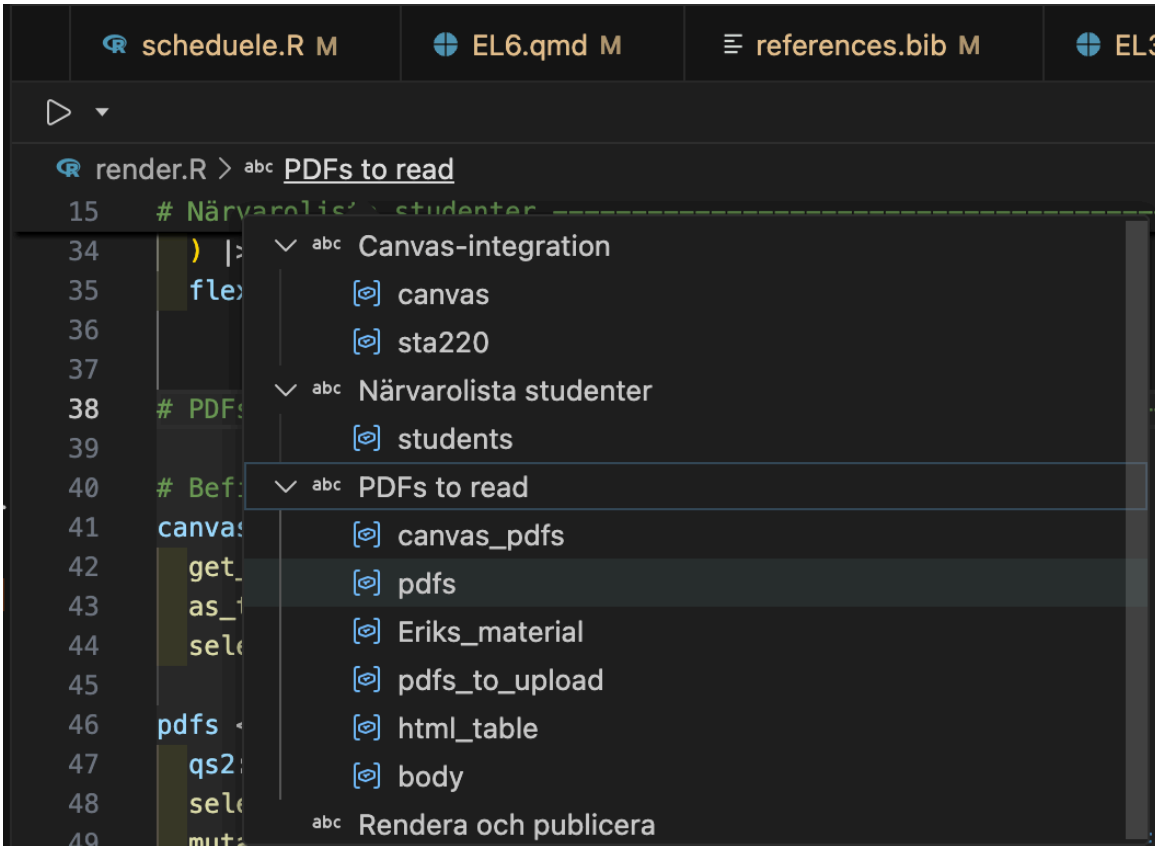
- **Code level**
  - scripts (comments, e.g. `# clean data and merge cohorts`)
  - functions (Roxygen2 documentation)
  - notes and reminders (e.g. `# TODO:`)
- **Project level**
  - `README.md`
  - (GitHub repos may also have a wiki or a polished website)
- **Output level**
  - figures and tables
  - reports (PDF / HTML)
  - presentations (slides)

Code level

Documenting R scripts

- Use Air for better formatting of the R code itself (will format the code according to best practice for readability and collaboration)
- Section labels by `Shift + Ctrl/Cmd + R` in Positron (and RStudio)
- Easiest way to aid navigation to R scripts

```
# My section heading -----------------------------------------------------
```



Individual scripts

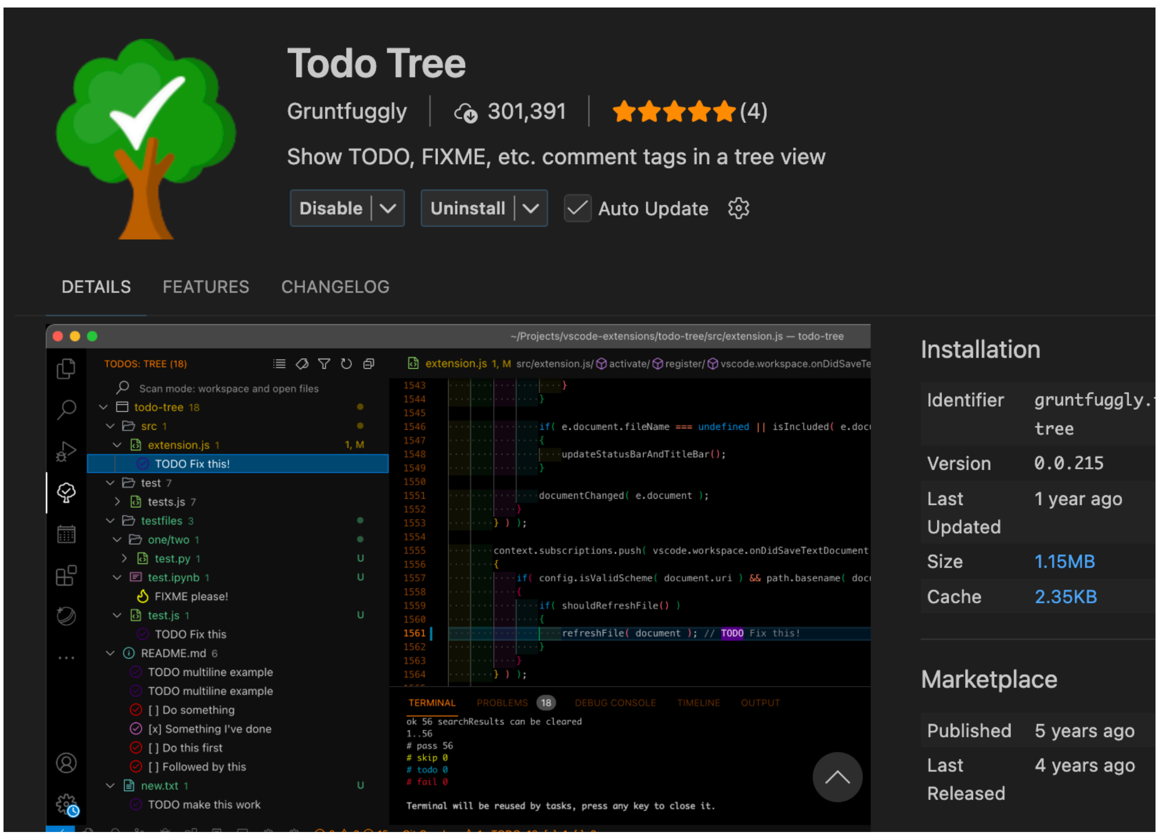
- As soon as you distribute an individual R script from a bigger project (by e-mail or as a printed copy) some context would get lost.
- The commentr package is designed to maintain such context
  - `remotes::install_bitbucket("cancercentrum/commentr")`

```
commentr::header_comment(
  "My nice script",
  "Bla bla bla ...",
  "Erik Bülow",
  "xxx@yyy.zzz",
  "John Doe"
)
```

```
The comment has been copied to clipboard and can be pasted into a script file!
```

```

###########################################################################
#                                                                         #
```

```
# Purpose:       My nice script                                         #
#                                                                       #
# Author:        Erik Bülow                                             #
# Contact:       xxx@yyy.zzz                                            #
# Client:        John Doe                                               #
#                                                                       #
# Code created:  2026-03-16                                             #
# Last updated:  2026-03-16                                             #
# Source:        /Users/xbuler/STA220/documentation                     #
#                                                                       #
# Comment:       Bla bla bla ...                                        #
#                                                                       #
#########################################################################
```




```
# TODO: Change this to something better:
1 + 2
```

Documenting R functions

What does this function do?

```r
hello <- function(who = "world") {
  sprintf("Hello %s!", who)
}
```

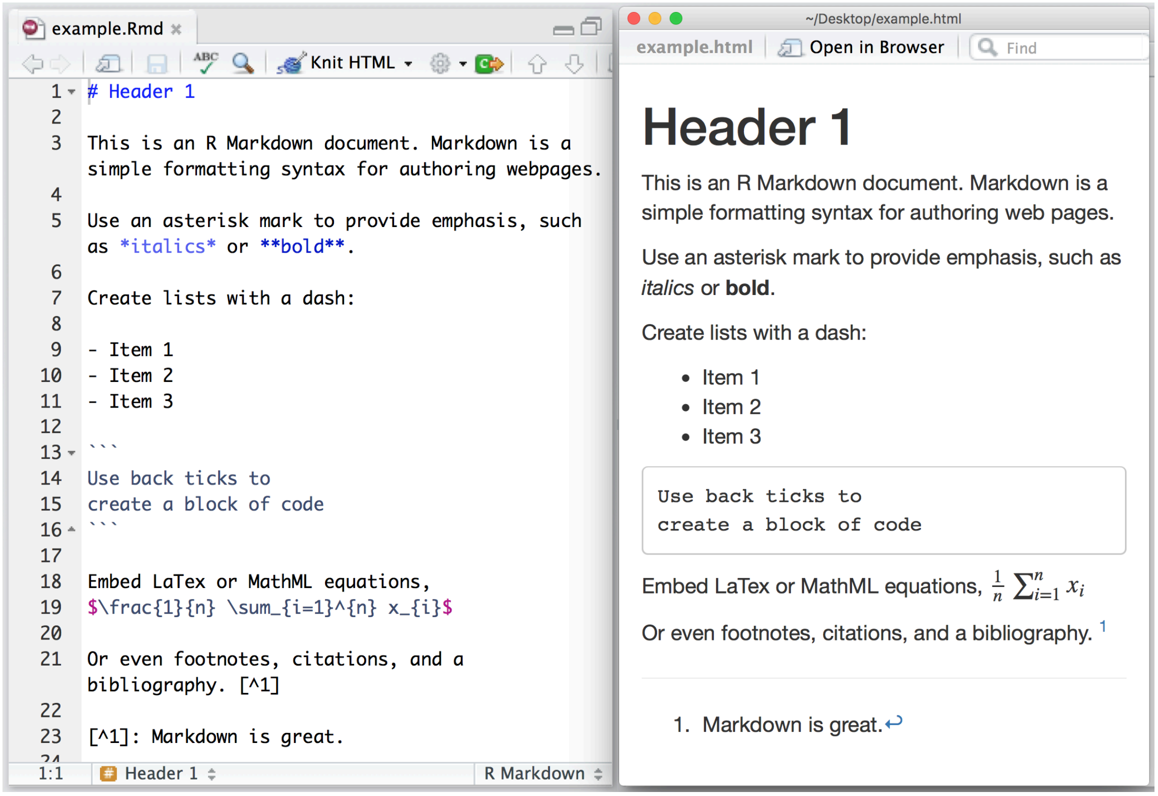
```r
#' Say hello
#'
#' Create a simple greeting. If no name is provided, the function
#' returns a greeting to the world.
#'
#' @param who A character vector giving the name(s) of the person or
#'   object to greet. If missing, the greeting defaults to `"world"`.
#'
#' @return A character vector of the same length as `who` containing
#'   greeting messages.
#'
#' @examples
#' hello()
#' hello("you")
#' hello(c("Alice", "Bob"))
hello <- function(who = "world") {
  sprintf("Hello %s!", who)
}
```

- There is a standardized syntax to document R functions
- You should be able to click a small light bulb (💡) to insert a sceleton when writing your function
- The same syntax is used within modern R packages
  - but it is useful even within your own projects
- roxygen2 is a package used for package documentation but it also has a great manual for documenteing functions in general

Project level

Documenting a project

- You have a `README.md` file within your git repository
- It can contain plain text without any formatting
- It will be displayed up-front in GitHub
- You can also include formatting!
- It will be nicely rendered on GitHub or elsewhere



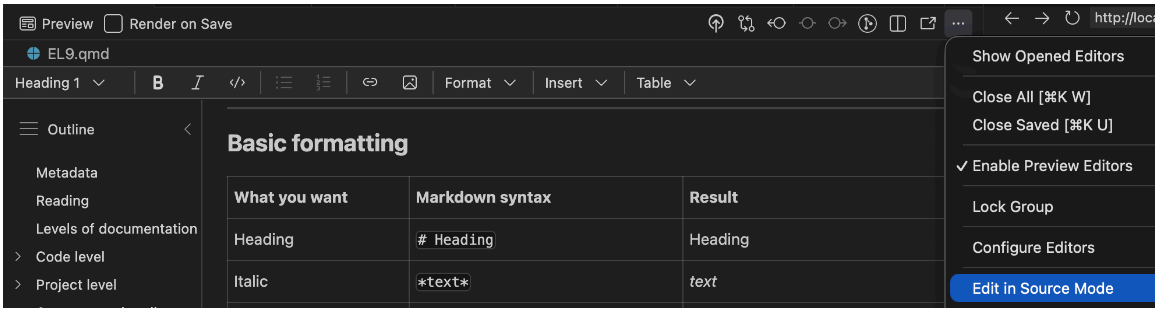
Basic formatting

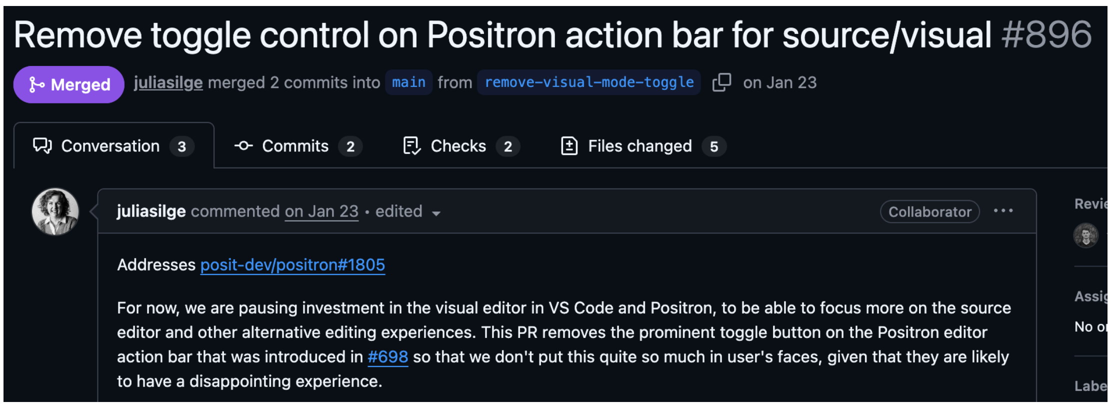
What you want	Markdown syntax	Result
Heading	`# Heading`	Heading
Italic	`*text*`	*text*
Bold	`**text**`	**text**
Inline code	`` `code` ``	`code`
Link	`[text](https://example.com)`	[text](https://example.com)
Bullet list	`- item`	• item
Numbered list	`1. item`	1. item

> **i Comment or heading**
>
> `#` is used for comments in R files and within R blocks but for headings in `.md` and `.qmd` files!

Visual mode

Instead of editing the source code, you may also use the “visual editor” in Positron:



> ⚠ Warning
>
> 
>

Use both?

Personally, I do most editing in source mode (better for the R chunks), but I find it much easier to include references/citations and to paste in external figures in using the visual mode.

Reproducible Workflows

Modern biostatistics projects rarely involve only statistics.

- data cleaning and preprocessing
- statistical modelling
- visualisation
- interpretation of results
- reporting and documentation

A good workflow should make it possible to **combine all of these steps in one place**.

Output level

What?

Quarto is a publishing system for scientific and technical documents.

It allows you to combine:

- text
- R code
- statistical results
- tables and figures

in a **single workflow**.

This makes it easier to create **reproducible statistical reports**.

Why?

In many projects people work with:

- R scripts for analysis 🥳
- Word documents for reporting 😱
- PowerPoint slides for presentations 😱

This separation often leads to:

- copy–paste errors 🤨
- outdated figures 👎
- results that cannot easily be reproduced 🤯

Quarto helps avoid these problems!

Reproducible analysis

In a Quarto document you can:

1. write the explanation of your analysis
2. include the R code used
3. generate figures and tables automatically
4. update everything by re-running the document

- If the data or model changes, the **entire report can be regenerated automatically**.
- It is widely used in **data science, epidemiology and biostatistics**.
- Integrated in Positron and developed by Posit PBC

Used in three ways

1. For **communicating to decision-makers**, who want to *focus on the conclusions, not the code behind the analysis.*
2. For **collaborating with collegues** (including future you!), who are interested in both your *conclusions, and how you reached them (i.e. the code).*
3. As an **environment** in which to do data science and statistics, as a modern-day lab notebook where you can capture not only what you did, but also what you were thinking.

A simple report

When you chose `New > New File ... > Quarto document`, the file will contain:

```yaml
---
title: "Untitled"
format: html
---
```

- This is a YAML header (Yet Another Markup Language)
- Different output formats are available:
  - **html:** can also be published online (Quartopub or GitHub pages)
  - **docx:** Word document for collaborating with non-statisticians etc
  - **typst:** PDF rendering (ever heard of $\LaTeX$? … This is the modern alternative!)
  - **revealjs:** slide decks (you are looking at one!)
- Can render to multiple formats at once!

This is the yaml header for this presentation:

```yaml
---
title: "EL9: Documentation"
subtitle: "Quarto"
reading: "[@wickham, ch. 28-29]"
format:
  revealjs:
    theme: blood
    transition: fade
    slide-number: true
    css: styles.css
    smaller: true
    scrollable: true
    incremental: false
  typst:
    include-before-body:
      - text: |
          // Gör alla länkar tydliga i PDF: blå + understrukna
          #show link: set text(fill: rgb("1a5fb4"))
          #show link: underline
  docx: default
bibliography: references.bib
---
```

YAML structure

YAML uses **indentation to represent structure**.

Think of it like a tree.

```yaml
format:
  html:
    code-fold: true
```

This corresponds to the structure:

```
format
└── html
    └── code-fold: true
```

If indentation is wrong?

```yaml
format:
html:
  code-fold: true
```

Now Quarto interprets this as:

```
format
html
└── code-fold: true
```

This is **not the intended structure**, and the document may fail to render.

💡 **Rule of thumb**

- use **two spaces per level**
- never use **tabs**
- when something fails, **check indentation first**

Three parts

Quarto documents have three parts:

1. An (optional) YAML header surrounded by `---`s.
2. Chunks of R code surrounded by ` ```{r} ` and ` ``` `.

````
```{r}          ← chunk header
#| echo: false  ← options controlling the output
summary(x)      ← R code
```             ← end of chunk
````

Keybindings: `Cmd + option + I` (Mac) or `Ctrl + Alt + I` (Windows)

3. Text with markdown format

Order

Part 1 (YAML header) always comes first but 2 (code block) and 3 (Markdown text) are usually intermingled throughout the rest of the document.

Inline code

Sometimes we want to insert **small pieces of R output directly in the text**.

This is called **inline code**.

Inline code will always just show the result of the call (not the code itself in the rendered document):

```
Todays date is `r Sys.Date()` and it is `r format(Sys.time(), format = "%H")`
```

> Todys date is 2026-03-16 and it is 09 AM

To higlight with bold and italic:

```
Todays date is **`r Sys.Date()`** and it is *`r format(Sys.time(), format =
"%I %p")`*
```

> Todays date is **2026-03-16** and it is *09 AM*

Preview Render on Save

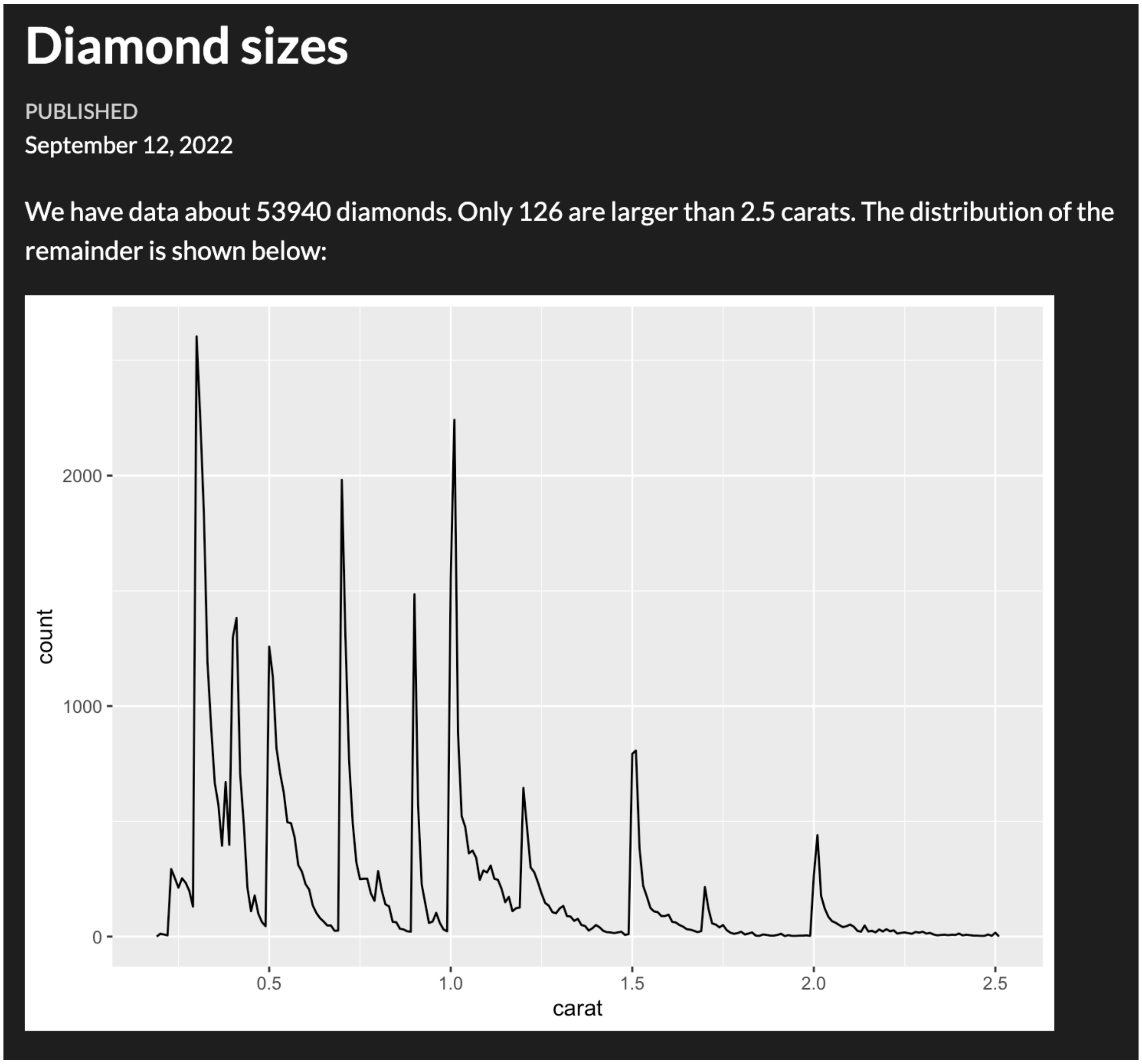
```
---
title: "Diamond sizes"
date: 2022-09-12
format: html
---

▷Run Cell | Run Next Cell
```{r}
#| label: setup
#| include: false

library(tidyverse)

smaller <- diamonds |>
  filter(carat <= 2.5)
```

We have data about `r nrow(diamonds)` diamonds.
Only `r nrow(diamonds) - nrow(smaller)` are larger than 2.5 carats.
The distribution of the remainder is shown below:

▷Run Cell | Run Above
```{r}
#| label: plot-smaller-diamonds
#| echo: false

smaller |>
  ggplot(aes(x = carat)) +
  geom_freqpoly(binwidth = 0.01)
```
```



To publish

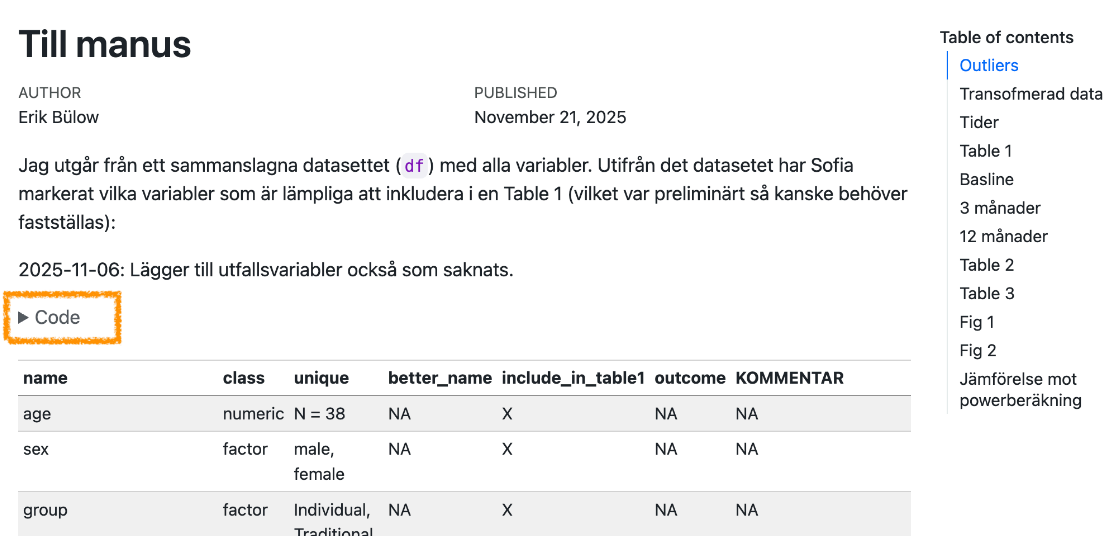
Decision makers, stake holders, friends or the public might not care about R. Disable R code (echoing the code itself), warnings and messages:

```
---
title: "Untitled"
format: html
execute:
  echo: false
  warning: false
  message: false
---
```

For collaborators

- Might still care less about verbose R messages and warnings (they trust that you have already considered those)
- May still need the underlaying R code for deeper understaing
- Use `code-fold`

```yaml
---
title: "Untitled"
format:
  html:
    code-fold: true
execute:
  warning: false
  message: false
---
```



Till manus

AUTHOR
Erik Bülow

PUBLISHED
November 21, 2025

Jag utgår från ett sammanslagna datasettet (`df`) med alla variabler. Utifrån det datasetet har Sofia markerat vilka variabler som är lämpliga att inkludera i en Table 1 (vilket var preliminärt så kanske behöver fastställas):

2025-11-06: Lägger till utfallsvariabler också som saknats.

▶ Code

name	class	unique	better_name	include_in_table1	outcome	KOMMENTAR
age	numeric	N = 38	NA	X	NA	NA
sex	factor	male, female	NA	X	NA	NA
group	factor	Individual, Traditional	NA	X	NA	NA

Table of contents

Till manus

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2025-11-06: Lägger till utfallsvariabler också som saknats.

▼Code

```
vars <- readxl::read_excel(here("docs/250529_vars_SM.xlsx"))
vars2incl <- vars$name[!is.na(vars$include_in_table1)]
vars2incl <- vars$name[!is.na(vars$include_in_table1) | !is.na(vars$outcome)]
vars |>
  filter(!is.na(include_in_table1) | !is.na(outcome)) |>
  knitr::kable()
```

name	class	unique	better_name	include_in_table1	outcome	KOMMENTAR
age	numeric	N = 38	NA	X	NA	NA
sex	factor	male, female	NA	X	NA	NA

Table of contents

This can also be set individually for each code chunk:

````
```{r}
#| echo: false
#| warning: false
#| message: true
message("will be muted!")
warning("will be ignored")
plot(something_beautiful, realy = TRUE)
```
````

Figures

Figures are usualy displayed without problem (and you can adjust their size, add captions etc):

````
```{r}
#| fig-cap: This is origo!
plot(0, 0)
```
````

```
plot(0, 0)
```

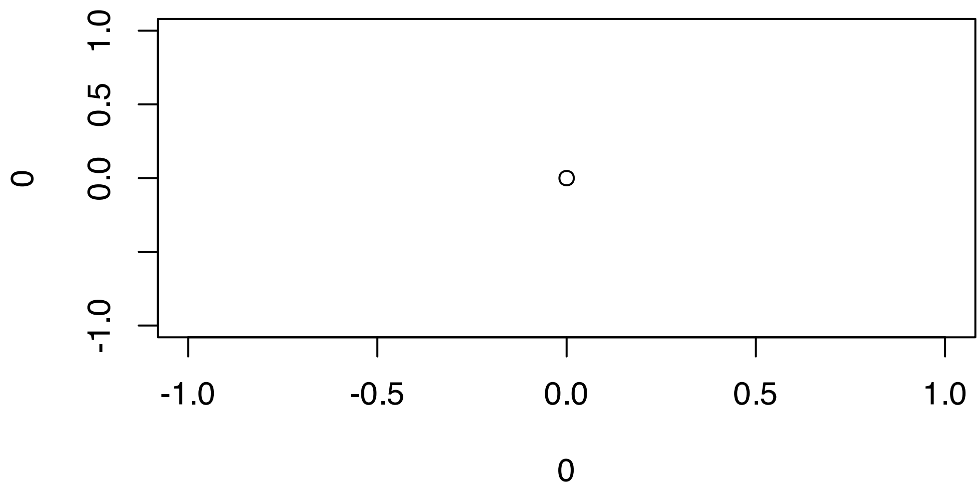


Figure 1: This is origo!

Tables

Just printing the R output might not look very nice:

```
head(iris)
```

```
  Sepal.Length Sepal.Width Petal.Length Petal.Width Species
1          5.1         3.5          1.4         0.2  setosa
2          4.9         3.0          1.4         0.2  setosa
3          4.7         3.2          1.3         0.2  setosa
4          4.6         3.1          1.5         0.2  setosa
5          5.0         3.6          1.4         0.2  setosa
6          5.4         3.9          1.7         0.4  setosa
```

Better with kable

The easiest way to make tibbles/data.frames/data.tables look nicer is with `df-print: kable` in the yaml header.

```
---
title: "Untitled"
format: html
df-print: kable
---
```

Or manually for individual outputs:

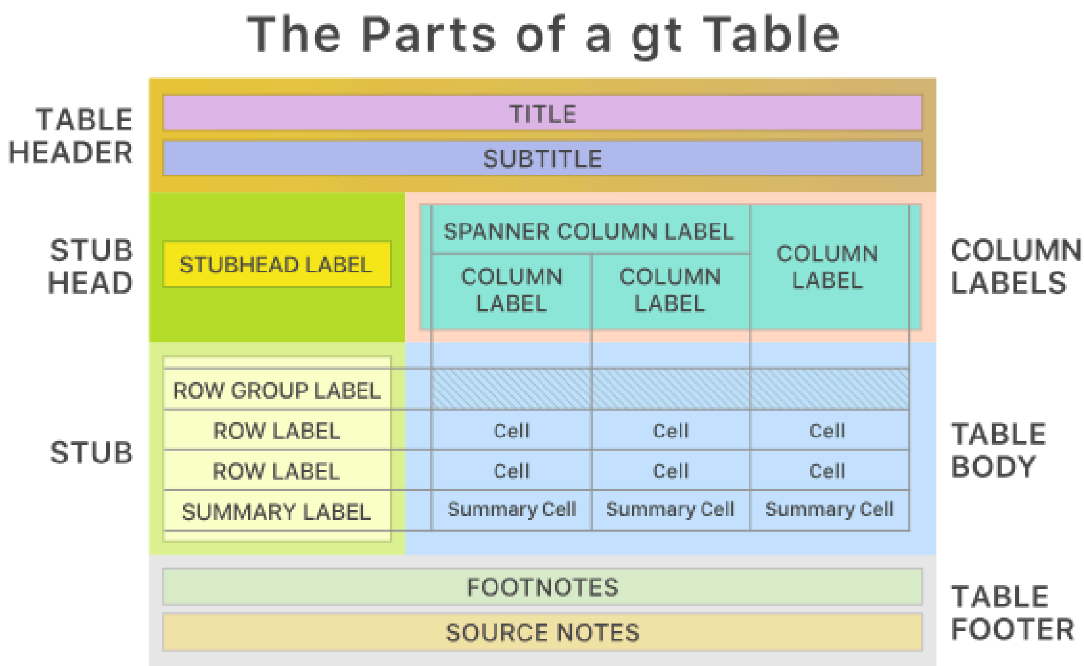
````
```{r}
#| tbl-cap: Some iris data
knitr::kable(head(iris))
```
````

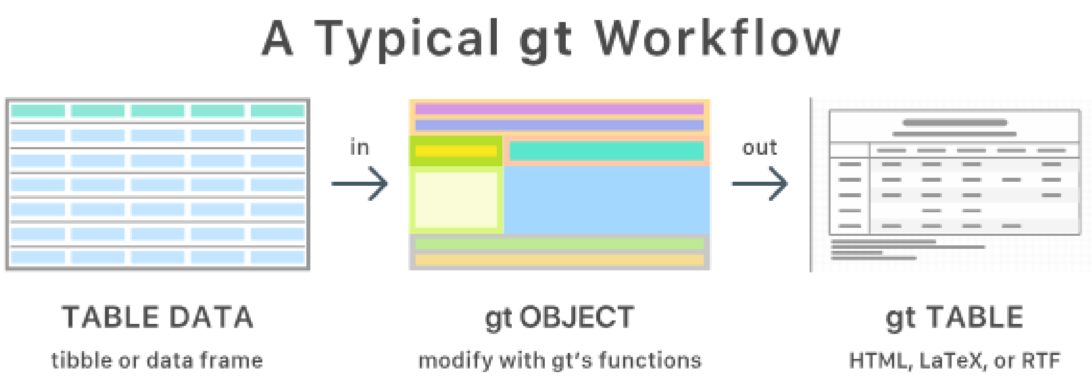
```
knitr::kable(head(iris))
```

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa

Best with gt

- Very flexible
- Publication ready
- Package from Posit
- flextable may be preferred instead if rendering Office documents (Word, PowerPoint)





gtsummary

gtsummary package build on top of {gt} to easily summarise datasets, regression models etc

```
```{r}
gtsummary::tbl_summary(iris, Species)
```
```

```
gtsummary::tbl_summary(iris, Species)
```

Characteristic	setosa N = 50[1]	versicolor N = 50[1]	virginica N = 50[1]
Sepal.Length	5.00 (4.80, 5.20)	5.90 (5.60, 6.30)	6.50 (6.20, 6.90)
Sepal.Width	3.40 (3.20, 3.70)	2.80 (2.50, 3.00)	3.00 (2.80, 3.20)
Petal.Length	1.50 (1.40, 1.60)	4.35 (4.00, 4.60)	5.55 (5.10, 5.90)
Petal.Width	0.20 (0.20, 0.30)	1.30 (1.20, 1.50)	2.00 (1.80, 2.30)

[1] Median (Q1, Q3)

Presentations

- Quarto can export to PowerPoint
- And to the older Beamer format
- But I recommend revealjs
  - `format: revealjs`
  - can also be modified with speaker notes, laser pointer etc
- Info and examples

For example (as used for these slides):

```
format:
  revealjs:
    theme: blood
    transition: fade
    slide-number: true
    css: styles.css
    smaller: true
    scrollable: true
    incremental: false
```

Integrate with {`targets`}

`tarchetypes::tar_quarto()`

in `_targets.R`

```
library(targets)
library(tarchetypes) # Let you integrate a quarto report/presentation
list(
  tar_target(data, data.frame(x = seq_len(26), y = letters))
  tar_quarto(report, "report.qmd") # Make presentation
)
```

In `report.qmd`

````
---
title: My report
format: html
---

# Here is my beautiful data!

```{r}
gt::gt(tar_read(data))
```
````

Try it

Copy this code to the terminal (not the console!) and try `targets::tar_make()` in the console!

```
cd
mkdir test_targets_quarto
cat > test_targets_quarto/_targets.R <<'EOF'
library(targets)
library(tarchetypes) # Lets you integrate a quarto report/presentation

list(
  tar_target(data, iris),
  tar_target(model, lm(Sepal.Length ~ ., data)),
  tar_quarto(report, "report.qmd") # Make presentation
)
EOF

cat > test_targets_quarto/report.qmd <<'EOF'
---
title: My project
author: My name
date: today
format:
  typst: default
  docx: default
  html: default
  revealjs:
    output-file: slides.html
---

## Here are some flowers!

gtsummary::tbl_summary(targets::tar_read(data))

## And here is a model

```

```
gtsummary::tbl_regression(targets::tar_read(model))
EOF

positron test_targets_quarto
# rm -r test_targets_quarto
```

Manuscripts

- Quarto Manuscript
- framework for writing and publishing scholarly articles
- Something for your later thesis work?
- Handles references
- Export to Word documents (`.docx`) etc
  - which is still the most common format for submission to medical/epidemiological journals
  - Still used for collaborative work with non-statisticians etc

Use references

When writing scientific reports we need to:

- acknowledge previous research
- support claims with evidence
- allow readers to find the original sources

This is done through **citations** and a **reference list**.

Example:

> Don't blame me, it was according to H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund [1].

At the end of the document a **bibliography** is automatically generated.

How Quarto handles references

Quarto uses a **bibliography file**, usually in `.bib` format.

Example in the document header:

```
bibliography: references.bib
```

Then citations can be written directly in the text:

```
Several studies have examined this relationship @smith2020.
```

Quarto will render something like:

> Several studies have examined this relationship (Smith 2020).

The full reference will appear automatically in the bibliography.

Online portfolio

- Looking for a job after you fininsh the master program?
- An online portfolia might increase your visability!
- Perhaps add your course project to such portfolio?
- Demo

Bibliography

[1] H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund, "R for Data Science (2e)." [Online]. Available: https://r4ds.hadley.nz/